Ujjwal Basumatary

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Education

Indian Institute of Science, Bangalore **Nov 2020 – Apr 2025**
Bachelor and Master of Science, Major in Physics *(Physics: 9.6/10) CGPA: 9.3/10*

Publications and Preprints

- arXiv:2410.22702 *Phys. Rev. D 111, L041306*
  Beyond Hawking evaporation of black holes formed by dark matter in compact stars, Ujjwal Basumatary, Nirmal Raj and Anupam Ray.
- arXiv:2603.26090
  Cosmological Correlators Using Tensor Networks, Ujjwal Basumatary, Aninda Sinha and Xinan Zhou.
- arXiv:2512.02706
  Lectures on Quantum Field Theory on a Quantum Computer, Aninda Sinha and Ujjwal Basumatary.
- *Thermalization and entanglement in the Yukawa–SYK model*. With Sumilan Banerjee. *Manuscript in preparation.*

Research Projects

Thermalization and entanglement in the Yukawa–SYK model **May 2025 – Present**
With Prof. Sumilan Banerjee *IISc, Bangalore*

- Developed a Schwinger–Keldysh treatment of the Yukawa–SYK model and numerically studied its large-N dynamics following oscillator-frequency quenches.
- Studying thermalization, Rényi entropies, and the relaxation of bosonic and fermionic correlations.

Tensor networks and quantum computing for QFT **Sep 2025 – Jun 2026**
With Prof. Aninda Sinha *IISc, Bangalore*

- Used matrix-product-state (MPS) methods to study cosmological correlators in quantum field theory; manuscript under review at the Journal of High Energy Physics.
- Coauthored lecture notes on QFT and quantum computing, with implementations of quantum and MPS algorithms for the associated teaching material. Notes are currently under review at Hindustan Book Agency.

Quantum-mechanical scattering on a photonic QC | Master's thesis **Sep 2024 – Apr 2025**
Supervisor: Prof. Aninda Sinha *IISc, Bangalore*

- Developed an arbitrary-precision package to simulate quantum-mechanical scattering in a photonic quantum-computing setting; applied it to a toy supermembrane model in my master's thesis.

Memory burden in black holes **Apr 2024 – Oct 2024**
Supervisor: Prof. Nirmal Raj *IISc, Bangalore*

- Derived and numerically solved evolution equations for memory-burdened black holes inside neutron stars using Mathematica code I developed.
- Extended exclusion bounds on dark matter–neutron scattering cross sections by several orders of magnitude; published in *Physical Review D*.

Conformal bootstrap | Bachelor's thesis **Sep 2023 – Apr 2024**
Supervisor: Prof. Aninda Sinha *IISc, Bangalore*

- Studied conformal dispersion relations and implemented Mathematica code for bootstrap calculations; used the Polyakov–Mellin bootstrap to study the ϵ-expansion in my bachelor's thesis.

Canonical gravity **May 2023 – Jul 2023**
Supervisor: Prof. Suvrat Raju *ICTS, Bangalore*

- Studied the Hamiltonian formulation of general relativity, de Sitter quantum gravity, and gauge-invariant cosmological perturbation theory; presented the work at the end of an eight-week reading project.

Lattice gauge theory **May 2022 – Jul 2022**
Supervisor: Prof. Anosh Joseph *IISER, Mohali*

- Implemented complex Langevin dynamics in C++ to compute observables for the bosonic degrees of freedom of the mini-BMN matrix quantum-mechanics model.

Employment

Lodha Theoretical Physics Institute, Mumbai **Jul 2026 – Present**
Research Assistant

Indian Institute of Science, Bangalore **Aug 2025 – Jun 2026**
Project Intern *Center for High Energy Physics*

Raman Research Institute, Bangalore **Aug 2025 – Jun 2026**
Project Intern, National Quantum Mission *Light and Matter Physics*

Indian Institute of Science, Bangalore **Jun 2025 – Jul 2025**
Project Associate - 1

Teaching Assistantships

HE381: Quantum Field Theory on a Quantum Computer **IISc, Bangalore**
Instructor: Prof. Aninda Sinha *CHEP*

- Led hands-on Qiskit tutorials and designed/graded assignments on quantum algorithms and Matrix Product States (MPS). The course led to a set of lecture notes.
- Implemented quantum and MPS algorithms for the course's `qftonqc` teaching codebase.

Fellowships and Awards

S. N. Bhatt Memorial Fellowship **May 2023 – Jul 2023**
International Center for Theoretical Sciences (ICTS) *Bangalore*

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship **Nov 2020 – Apr 2025**
Department of Science and Technology (DST) *India*

National Talent Search Exam (NTSE) Scholarship **Apr 2018 – Mar 2020**
National Council of Educational Research & Training (NCERT) *India*

Skills

- **Programming:** C, C++, Python, shell script, Mathematica, Julia and Haskell.
- **Markup:** Proficient in LaTeX, working knowledge of HTML and CSS.
- **OS:** Windows, GNU/Linux.

Advanced Graduate Coursework

Non-eq. Many-Body Dynamics	10/10	AdS_3/CFT_2	9/10
Standard Model	10/10	Adv. Mathematical Physics	10/10
QFT - 2	9/10	Cosmology for Theorists	9/10
Statistical Field Theory	9/10	QFT - 1	10/10
General Relativity	10/10	Nuclear and Particle Physics	10/10